Academy

AI & Automation Curriculum

Become job-ready in 3-6 months — Learn. Build. Excel.

Scholarships available

Course Overview

This course introduces students to Artificial Intelligence (AI) and Automation fundamentals, exploring how intelligent systems are built, trained, and deployed to solve real-world problems. You'll learn how to use AI tools, automation frameworks, and no-code platforms to streamline workflows, build intelligent systems, and drive efficiency across industries.

Designed for beginners and professionals alike, this program blends theory, hands-on projects, and real business applications to help you gain the skills needed to thrive in the age of intelligent technology.

Learning Model

Mode

Online Live Classes or Self Paced with Recording (You Choose)

Duration

3 months

Level

Beginner to Intermediate

Accreditation

ACTD



Learning Outcome

By the end of this course, you will be able to:

1. Understand the core principles of Artificial Intelligence, Machine Learning, and Automation.
2. Build simple automation workflows using no-code and low-code tools such as Zapier, Make, N8N and ChatGPT API.
3. Apply AI tools for business operations, marketing, analytics, and productivity enhancement.
4. Design and deploy automation-driven solutions that improve efficiency and decision-making.

Career Opportunities



United States
$85,000 – $120,000 USD



United Kingdom
£50,000 – £75,000 GBP



Canada
$70,000 – $95,000 CAD



Germany
€55,000 – €80,000 EUR



Nigeria
~~₦~~5.5M – ~~₦~~9M NGN



Kenya
KSh 2.2M – KSh 3.8M



Remote/Global Roles
$60,000 – $100,000 USD

Modules

Module 1

Introduction AI & Automation

- What is AI?
- What is Prompt Engineering?
- What is Automation?
- Types of Automation
- Overview of Automation Platforms
- Relationships & Advanced Data Structures
- Examples & practice questions

Module 2

AI Automation in Practice

- Planning an Automation Project
- Automation Best Practices
- Workflow Design Principles
- Common Automation Use Cases

Modules

Module 3

Zapier & Smart Workflow Automation

- Introduction to Zapier and how it works
- Triggers, Actions, and Paths
- Formatters and Delays
- Webhooks
- Code by Zapier
- Integrating Google Workspace tools

Module 4

Make & Advanced Workflow Automation

- Introduction to Make and how it works
- Scenarios, Modules, and Data Mapping
- Triggers and Actions
- Routers, Filters, and Logic
- Webhooks and Custom Integrations
- Data Stores and API Connections
- Building Multi-App Workflows
- Real-World Case Studies and Assignments

Modules

Module 5

n8n Advanced Workflow Automation

- Introduction to n8n and its interface
- Understanding Nodes, Triggers, and Workflows
- Building simple automations step by step
- Using Logic Controls and Data Transformations
- Webhooks and API Integrations
- Error Handling and Execution Control
- Practice: Create a 3-step automated workflow

Module 6

Essential n8n Nodes & Core Building Blocks

- The 10 Essential Nodes at a Glance
- Logic Nodes
- If Node – Binary yes/no decision gate
- Switch Node – Multi-path routing based on rules
- Merge Node – Combines two or more data streams into one
- Data Nodes
- Set Node – Add, remove, or rename data fields
- Split in Batches Node – Divides large lists into smaller groups
- Aggregate Node – Collects multiple items into a single output
- Control Nodes
- HTTP Request Node – Connect to any external API or web service
- Code Node – Run custom JavaScript or Python logic
- Wait Node – Pause the workflow for a set duration
- Error Trigger Node – Catch and handle errors from other nodes
- Choosing the Right Node for the Job
- Practice Exercises: Node Identification and Configuration

Modules

Module 7

Combining Nodes to Build Real Workflows

- Introduction: From Individual Nodes to Complete Systems
- Workflow Architecture and Node Selection Strategy
- Complete Workflow 1: Lead Qualification System
- Webhook → Set → If → Switch → HTTP Request → Merge
- Complete Workflow 2: Order Processing Pipeline
- Trigger → Split in Batches → Code → If → HTTP Request → Aggregate → Error Trigger • Complete Workflow 3: Customer Support Ticket Router
- Webhook → Set → Switch → Parallel Paths → Merge → HTTP Request
- Debugging and Testing Multi-Node Workflows
- Best Practices for Production Workflows
- Hands-On Exercise: Build a Complete End-to-End Workflow

Module 8

Chatbot Development & Implementation

- Introduction to Chatbots
- Chatbot Use Cases
- Chatbot Building Platforms(Voiceflow, n8n, Chatbase, ManyChat)
- Other Platforms
- Advanced Chatbot Features
- Exercise

Modules

Module 9

AI Agents & Agentic Systems

- Understanding AI Agents
- Types of AI Agents
- Low-Code/No-Code AI Agent Platforms(n8n, Relevance AI)
- Code-First Agentic AI Frameworks(Langchain)
- Multi-Agent Systems
- AI Agent Capabilities
- Real-World AI Agent Applications
- Governance and Safety
- Exercise 1: Design an AI agent workflow for a specific business task
- Exercise 2: Build a simple AI agent using a no-code platform (n8n, Lindy, Flowise, or Voiceflow)

Module 10

AI Agents & Smart Automation with n8n

- Introduction to AI-powered automation
- Connecting n8n with AI tools (OpenAI, ChatGPT, Claude, etc.)
- Using APIs to build AI Agents
- Integrating tools like ManyChat, Voiceflow for conversational automation
- Combining AI with data sources (Airtable, Google Sheets, etc.)
- Automating tasks for lead generation, and customer support
- Building multi-channel workflows across email, chat, and web forms
- Case Study: Build a basic AI Agent workflow
- Assignment: Create an AI-powered task automation

Modules

Module 11

Vibe Coding & AI-Powered Development

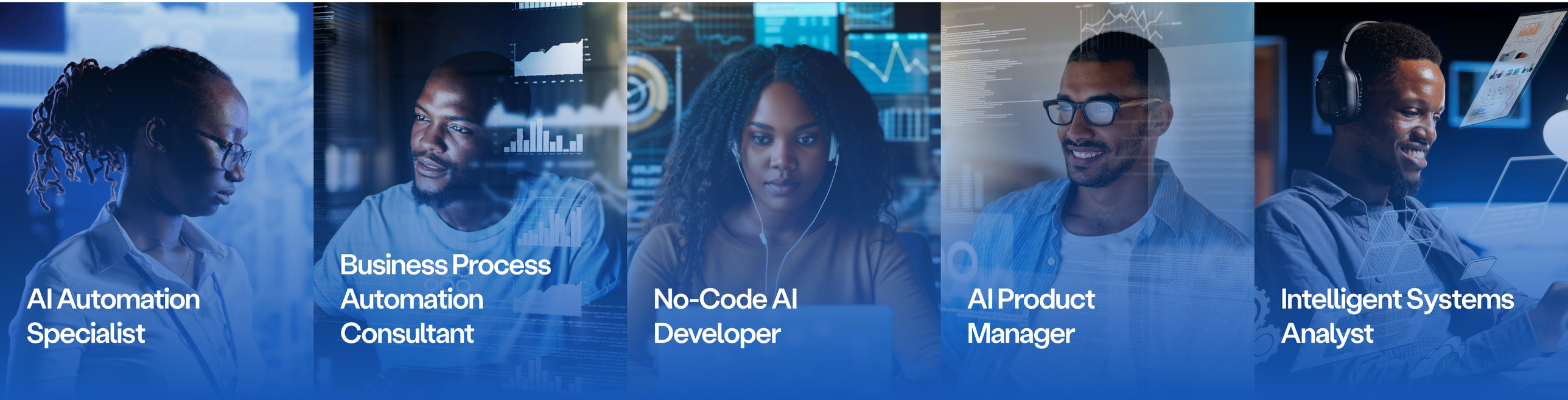
- Introduction to Vibe Coding
- The Vibe Coding Philosophy
- AI-Powered IDEs and Code Editors
- No-Code/Low-Code AI Development Platforms(Bolt.new,
- Vibe Coding Workflow
- Challenges and Limitations
- Exercise

Module 12

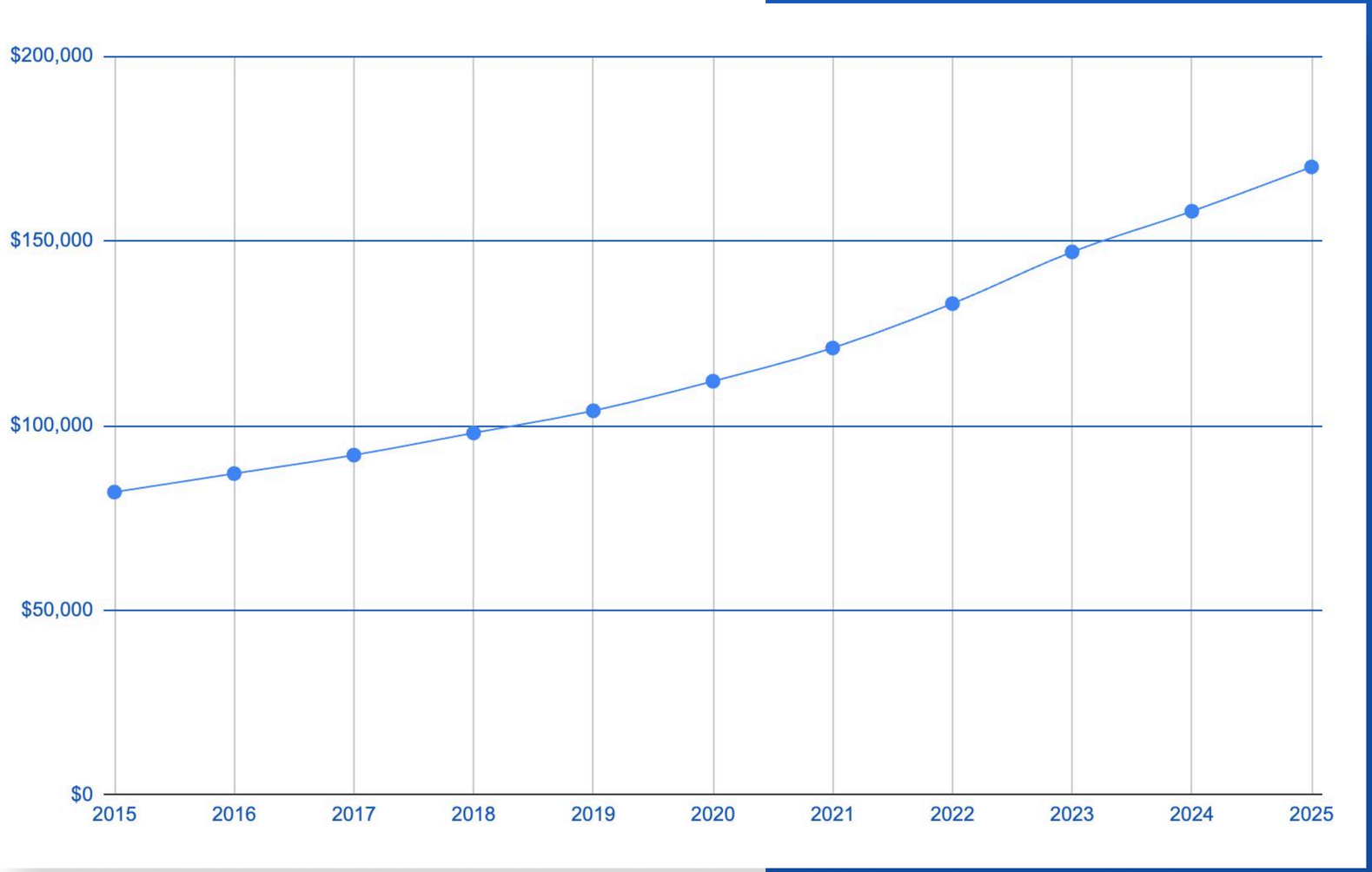
AI Voice Agents & Conversational AI

- Introduction to AI Voice Agents
- Voice Agent Capabilities
- Platforms for Building AI Voice Agents(Vapi, Retell AI)
- Other Voice AI Platforms(ElevenLabs)
- Voice Agent Use Cases
- Best Practices for Voice AI Agent
- Exercise

Career Pathways



10 Years Salary Trend



Why Choose TS Academy

- Scholarships Available
- Beginner-Friendly Curriculum
- Live + Self-Paced Learning Options
- Hands-On Practical Projects
- Mentorship & Accountability Support
- Internship & Job Placement Assistance



Get in Touch

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